

Stony Brook University
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Fall 2016

MIDTERM III

Due: Monday December 12, 2016, before 4:00PM

INSTRUCTIONS: This is a take-home examination. You are *encouraged* to discuss with your classmates and work in groups, however, *each student must deliver her or his own solutions*. Since this is a take-home examination, I will be very careful when grading, so please provide *complete and detailed answers*. Your answers should also be neatly written. This examination is due on December 12, before 4:00PM in my department mailbox, located in the 6th floor of the SBS building near the coordinator's office. I *will not accept any late exams or online submissions* so please plan ahead so you can deliver the hard-copy on time.

1. Suppose that the utility function for goods x and y is given by:

$$U(x, y) = (x + 1)y.$$

- (a) Given a fixed level of utility, $\bar{U}$, graph the indifference curve. Based on your graph, explain why you might get a corner solution.
 - (b) Calculate the uncompensated (Marshallian) demand functions for x and y . Be careful of the possibility for having a corner solution for x , i.e. $x^* = 0$ and explain what conditions need to hold so that you get a positive solution for x^* .¹
 - (c) Describe how the Marshallian demand curve for x and y are shifted by changes in I or in the price of the other good.
 - (d) Calculate the expenditure function for x and y .
 - (e) Use the expenditure function in the previous part to compute the compensated demand functions for goods x and y . Describe how the compensated demand curves for x and y are shifted by changes in income or by changes in the price of the other good.
2. Suppose that Laura regards ham and cheese as pure complements—she will always use one slice of ham in combination with one slice of cheese to make a ham and cheese sandwich. Suppose also the ham and cheese are the only goods Laura buys and that bread is free.
- (a) If the price of ham is equal to the price of cheese, show that the own-price elasticity of demand for ham is -0.5 and that the cross price elasticity of demand for ham with respect to the price of cheese is also -0.5 .
 - (b) Explain why the results from the previous part reflect only income effects, not substitution effects. What are the compensated price elasticities in this problem?
 - (c) Use the results from the previous part to show how your answers in part (a) would change if a slice of ham cost twice the price of a slice of cheese.
 - (d) Explain how this problem could be solved intuitively by assuming Laura consumes only one good—a ham and cheese sandwich.

3. Consider a simple quasi-linear utility function of the form $U(x, y) = x + \ln(y)$.²

¹At a corner solution, you won't necessarily have that $MRS = p_x/p_y$. Hint: Assume that the tangency condition holds and solve for the optimal demands. Then see what condition needs to hold for these demands to make sense. If this condition doesn't hold, then you get the corner solution.

²Check the solutions for homework 2.

- (a) Calculate the income effect for each good. Also calculate the income elasticity of demand for each good.
 - (b) Calculate the substitution effect for each good. Also calculate the compensated own-price elasticity of demand for each good.
 - (c) Show that the Slutsky equation apply to this function.
 - (d) Show that the elasticity form of the Slutsky equation also applies to this function. Describe any special features you observe.
4. There are two goods, food and clothing, whose quantities are denoted by x and y and prices p_x and p_y respectively. There is a consumer whose income is denoted by I and utility by U . His expenditure function is known to be ³

$$I = E(p_x, p_y, U) = \sqrt{p_x p_y} U.$$

- (a) Find this consumer compensated and uncompensated demand functions. You should not derive them by first principles, but instead by using some of the results we saw in class.
 - (b) Initially $I = 100$, $p_x = 1$ and $p_y = 1$. What quantities does the consumer buy, and what is his resulting utility?
 - (c) Suppose that there is an increase in the price of food caused by the government levying a tax of 21% on it. How much will this consumer buy now and what is his resulting utility? What is the government revenue from this tax?
 - (d) What is the welfare loss from this tax? Answer this question by calculating the compensating variation.
 - (e) Consider the government revenue you calculated in part (c) and denote it by R^* . Suppose now that instead of levying a tax of 21% on food, the government decides to tax this individual via an income tax in the amount of R^* . This means that the consumer will have an after tax income of $I - R^* = 100 - R^*$, while facing prices of $p_x = p_y = 1$ and the government will still collect R^* in revenues. What are the consumer's optimal choices and his resulting utility?
 - (f) Based on your answers to (e), is income taxation better than ad valorem taxation? Give the economic intuition behind this result. Ideally, you should argue your case by appealing to the Slutsky equation and the income and substitution effects.
5. The compensated and the uncompensated demand, i.e. the Marshallian and Hicksian demand function, can be related through the Slutsky equation. The Slutsky equation can be derived from the equality

$$x(p_x, p_y, E(p_x, p_y, U)) = x^c(p_x, p_y, U). \quad (1)$$

- (a) State the Slutsky equation and briefly explain all the terms appearing in it.
- (b) Formally derive the Slutsky equation using equation (1).
- (c) Graphically illustrate the content that the Slutsky equation holds, carefully labeling all the objects that enter the graphic.

³You have met this problem before under a different guise, but now you have more tools to analyze it.